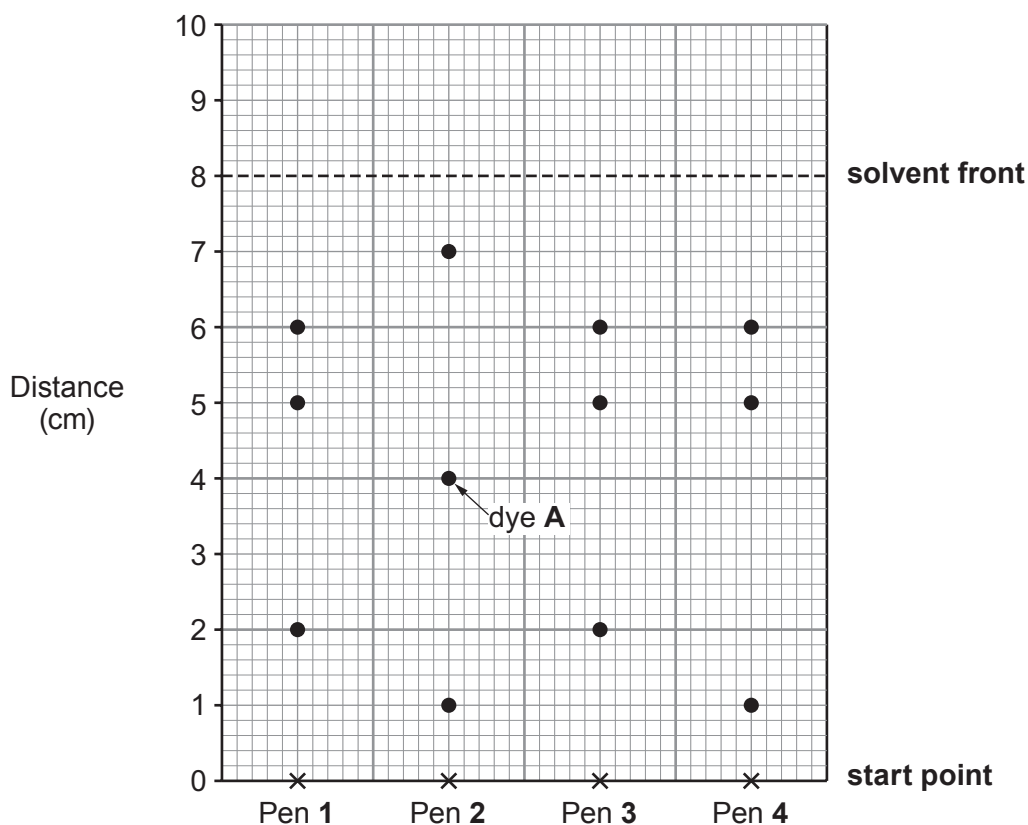


- (b) A group of students were asked to investigate the dyes in four different water soluble blue pens, **1**, **2**, **3** and **4**. The results are shown below.



- (i) Which **two** pens contain the same three dyes? [1]
Pens and
- (ii) Which pen contains the **most** soluble dye? [1]
Pen
- (iii) The R_f value of a substance can be used to identify that substance.
The R_f value is given by the formula:

$$R_f = \frac{\text{distance moved by the substance}}{\text{distance moved by the solvent front}}$$

Calculate the R_f value for dye **A**. [2]

$R_f =$

- (iv) The box contains methods for separating mixtures.

distillation	chromatography	filtration
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Complete the sentence using a method from the box.

The method used to separate the coloured dyes in the blue pens is called

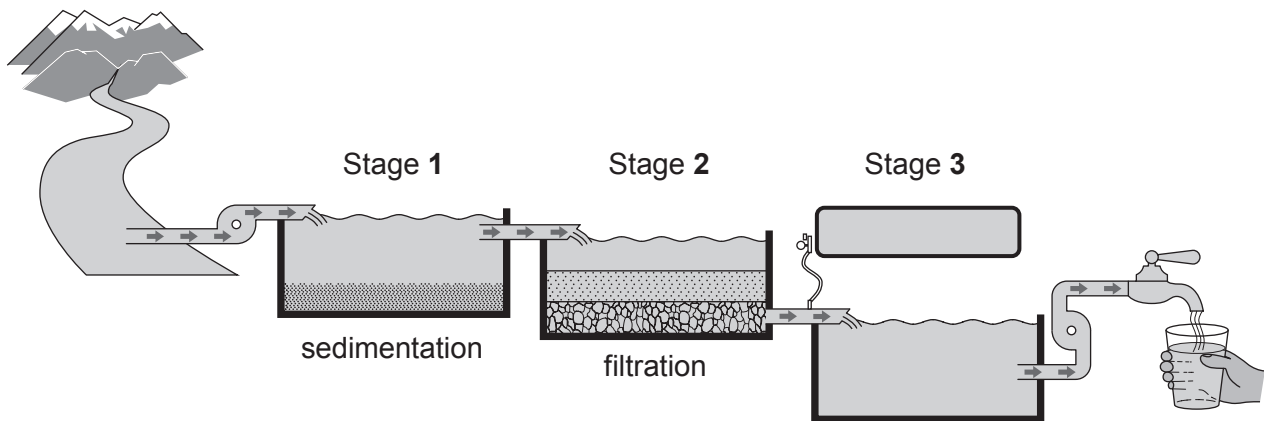
.....

[1]

8

4. (a) Most water in Wales is sourced from surface water and ground water. Ground water is the water beneath the surface of the Earth, consisting mainly of surface water that has filtered through the rocks. It is the source of water found in springs and wells. Human activity causes surface water to be more contaminated than ground water.

The diagram below shows the three main stages in the treatment of surface water before it enters the mains water system on its way to the taps in your home.



- (i) Tick (✓) the box that describes what occurs in stage 1. [1]

- | | |
|--|--------------------------|
| large insoluble particles sink to the bottom | ☐ |
| soluble particles are removed | ☐ |
| small fine insoluble particles are removed | ☐ |
| fluoride is added to the water | ☐ |

- (ii) Stage 3 makes water safe to drink. Name the substance added to water in stage 3. State why this makes the water safe to drink. [2]

Substance

- (iii) Describe **one major** cause of surface water pollution. [2]

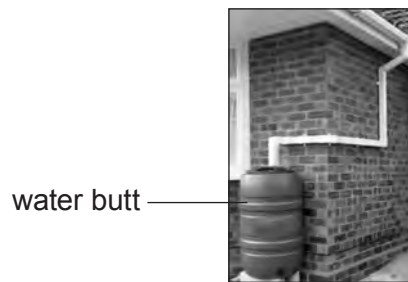
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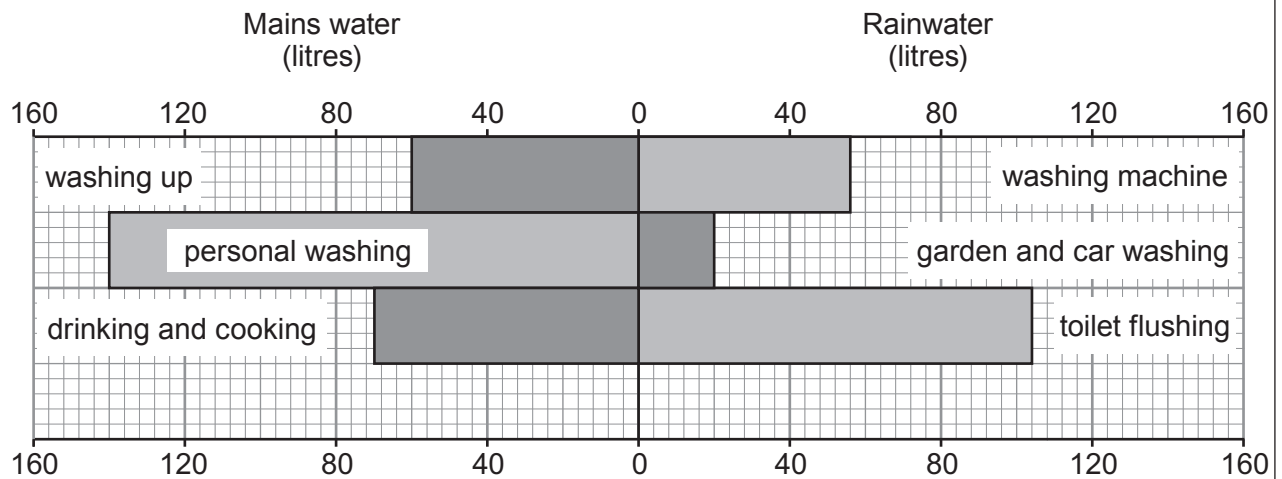
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- (b) A family uses a water butt to collect rainwater in order to reduce their use of mains water.



They use a total of 450 litres of water on a given day. The chart shows the volumes of mains water and collected rainwater used for various purposes on that day.



- (i) What percentage of the total water used on that day is rainwater? [2]

percentage rainwater = %

- (ii) Suggest why it is safe to use rainwater for the uses shown on the right hand side of the chart. [1]

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